

LYNEHAM, United Kingdom

WMO: 037400

Lat: 51.5031N

Lon: 1.9922W

Elev: 476

StdP: 14.45

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.4	27.8	20.8	15.9	29.0	23.2	17.8	30.9	30.3	45.8	27.4	45.1	8.5	30	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	78.7	64.6	75.0	62.9	71.8	61.3	66.2	75.0	64.4	71.8	62.8	69.2	9.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.1	87.9	68.8	61.6	83.3	66.7	60.2	79.2	65.1	31.0	75.0	29.7	72.0	28.5	69.3	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.6	21.7	19.3	DB	21.4	84.3	3.3	4.0	19.0	87.1	17.1	89.4	15.3	91.7	12.9	94.5
			WB	20.7	69.9	3.2	2.7	18.4	71.9	16.6	73.5	14.8	75.0	12.5	77.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	39.9	40.5	43.6	47.6	53.0	58.3	62.2	61.6	57.4	51.8	45.1	40.8
	DBStd	9.50	5.84	5.63	5.14	5.16	5.15	4.53	4.49	4.26	4.09	4.92	5.50	6.50
	HDD50	1402	313	267	205	104	25	0	0	0	1	37	162	288
	HDD65	5458	777	686	665	523	372	207	113	126	232	411	597	749
	CDD50	1477	1	1	5	31	119	251	378	358	222	92	15	4
	CDD65	56	0	0	0	0	1	7	26	20	2	0	0	0
	CDH74	420	0	0	0	1	8	67	204	126	12	2	0	0
	CDH80	67	0	0	0	0	0	9	34	24	0	0	0	0
Wind	WSAvg	9.7	10.9	10.8	10.7	10.0	9.8	8.9	8.4	8.3	8.8	9.5	9.6	10.3
Precipitation	PrecAvg	28.1	2.7	1.9	2.1	2.0	2.2	2.2	2.0	2.3	2.5	2.6	2.7	3.0
	PrecMax	38.9	5.2	5.5	5.9	6.0	5.5	5.7	4.6	5.3	5.6	6.1	5.6	5.9
	PrecMin	18.8	0.3	0.2	0.1	0.0	0.2	0.1	0.2	0.1	0.4	0.2	1.0	0.6
	PrecStd	4.3	1.4	1.3	1.1	1.1	1.2	1.5	1.1	1.4	1.5	1.5	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	55.1	61.2	69.8	74.9	80.7	84.1	83.1	75.9	67.2	58.6	55.5
		MCWB	52.0	50.3	51.5	57.6	61.1	64.8	65.7	65.4	63.4	60.8	56.2	53.5
	2%	DB	52.2	51.9	56.5	63.6	70.5	75.1	79.5	77.3	70.7	62.7	56.6	53.8
		MCWB	50.4	48.8	49.7	53.7	59.3	63.2	64.6	64.6	61.4	58.1	54.5	52.3
	5%	DB	50.2	50.1	53.6	59.3	66.2	71.2	75.5	72.9	67.3	60.7	54.9	52.2
		MCWB	48.5	47.8	48.5	51.3	57.3	60.7	62.7	62.4	59.4	57.0	52.8	50.5
	10%	DB	48.5	48.6	51.3	56.1	62.3	67.4	71.8	69.5	64.8	59.0	53.3	50.4
		MCWB	46.8	46.4	47.4	49.6	55.1	58.7	61.5	61.0	58.2	55.9	51.4	48.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	51.8	53.1	58.6	63.6	68.0	68.2	68.8	65.6	61.9	57.2	54.0
		MCDB	53.4	54.0	57.4	66.7	71.7	77.6	79.3	78.7	72.3	65.8	58.0	54.7
	2%	WB	50.8	49.8	51.2	54.8	60.5	64.0	65.9	65.9	63.0	59.5	54.9	52.7
		MCDB	51.8	51.3	54.6	61.9	67.5	72.0	75.8	73.2	67.9	61.7	56.0	53.6
	5%	WB	48.7	48.3	49.8	52.6	58.3	61.8	64.3	64.0	61.2	58.0	53.3	51.0
		MCDB	49.9	49.8	52.6	57.7	64.8	68.9	72.9	70.0	65.4	60.0	54.6	52.1
	10%	WB	47.0	46.7	48.3	50.6	56.2	59.9	62.5	62.4	59.6	56.6	51.5	49.0
		MCDB	48.3	48.5	50.8	55.2	61.5	66.1	69.5	67.8	63.4	58.5	53.1	50.3
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.7	11.9	14.3	14.7	15.0	15.5	14.5	13.7	11.2	9.3	8.5
		MCDBR	9.9	10.2	14.9	18.9	20.3	21.2	22.1	19.5	16.9	11.9	9.3	9.0
		MCWBR	9.6	8.4	9.8	11.1	11.3	11.2	10.0	9.2	9.0	8.2	8.2	9.0
	5% WB	MCDBR	9.9	9.9	12.9	16.7	18.1	18.3	19.0	16.4	14.7	10.9	9.3	9.1
		MCWBR	10.0	9.0	9.6	10.7	10.8	10.8	10.0	9.1	9.4	8.7	8.7	9.3
Clear-Sky Solar Irradiance	taub	0.321	0.336	0.370	0.391	0.395	0.390	0.386	0.376	0.362	0.348	0.322	0.311	
	taud	2.363	2.381	2.285	2.239	2.262	2.304	2.322	2.373	2.404	2.425	2.411	2.341	
	Ebn at Noon	217	249	260	268	272	274	273	271	261	241	216	200	
	Edh at Noon	20	25	34	39	41	39	38	34	30	24	19	17	
All-Sky Solar Radiation	RadAvg	253	466	817	1258	1536	1683	1584	1315	991	569	303	198	
	RadStd	22	59	108	146	88	176	154	112	79	51	32	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+1.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (83 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page